

Problem Set 5. Linear Regression

Quantitative Methods for Humanities

1. University Life and Final Grade.

A university wants to study whether weekly hours spent in student associations are related to students' final grade in an Economics course. For a sample of students, the following simple linear regression is estimated:

$$\widehat{\text{grade}} = 5.8 + 0.22 \text{association_hours}.$$

Grades are measured from 0 to 10, and **association_hours** measures hours per week.

- (a) Interpret the intercept and the slope coefficient in this context.
- (b) A student who spends 18 hours per week in student associations obtains a final grade of 9.44. Does the linear model under or overestimate the student's performance in the final exam?
- (c) The observed values of **association_hours** in the sample range from 0 to 12. Was the prediction for the student in the previous part reliable? Explain.

2. Predicting Support for Rent Control Policies.

A political scientist studies whether citizens who spend a larger share of their income on rent are more likely to support rent control policies. The variable **rent_burden** measures the percentage of monthly income spent on rent. The variable **urban** is a dummy equal to 1 if the respondent lives in a large city and 0 otherwise.

The researcher estimates the following model using survey data:

$$\widehat{\text{support}} = 22 + 0.85 \text{rent_burden} + 9 \text{urban}.$$

Support for rent control is measured on a 0–100 scale. The sample used to estimate the model contains values of **rent_burden** between 10 and 55.

- (a) Interpret the coefficient on **rent_burden**.
- (b) Interpret the coefficient on **urban**.
- (c) What is the predicted difference between the a non-urban and an urban respondent who spend the same percentage of their income on rent?
- (d) A new respondent spends 65% of their income on rent. Is this prediction within range or outside range? Why does this matter?

- (e) The researcher worries that the relationship between rent burden and support for rent control is not linear. Write a transformed linear model that allows rent burden to have a nonlinear effect while remaining a linear regression model in the parameters.

3. Civic Education Workshops: Regression and Difference in Means.

A political science department runs a randomized experiment to study whether civic education workshops increase citizens' political knowledge. Some citizens are randomly assigned to attend a short workshop about local institutions, while others are not invited. Let Y be the number of correct answers in a political knowledge quiz for a citizen, and let $T = 1$ if the citizen was assigned to the workshop and $T = 0$ otherwise.

The data are summarized as follows:

Group	Number of citizens	Mean of Y	Mean baseline interest in politics
Control, $T = 0$	120	5.2	3.0
Treatment, $T = 1$	125	6.1	3.1

- (a) Consider the regression $Y = \beta_0 + \beta_1 T + u$. What are the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$?
 (b) Interpret $\hat{\beta}_1$ as a difference in means. Under random assignment, what is its causal interpretation?
 (c) Why is random assignment important for this causal interpretation?
 (d) A second researcher estimates

$$Y = \beta_0 + \beta_1 T + \beta_2 \text{interest} + \beta_3 (T \cdot \text{interest}) + u.$$

Interpret β_3 .

- (e) According to the interaction model, what is the treatment effect for a citizen with interest = 4? Give the formula.

4. Scholarships and a Regression Discontinuity Design.

A regional government awards a university scholarship to students who score at least 80 points on an entrance exam. Let X be the exam score, let $T = 1$ if $X \geq 80$, and let Y be first-year university enrollment, measured as 1 if the student enrolls and 0 otherwise.

The researcher estimates the following regression:

$$Y = \beta_0 + \beta_1 X + \beta_2 T + \beta_3 (T \cdot X) + u.$$

The estimated coefficients are:

$$\hat{\beta}_0 = -0.55, \quad \hat{\beta}_1 = 0.015, \quad \hat{\beta}_2 = 0.30, \quad \hat{\beta}_3 = -0.001.$$

- (a) Compute the estimated jump at the cutoff $c = 80$.
 (b) State the key assumption that makes this jump a valid estimate of the average treatment effect.
 (c) Explain why this RDD estimate may have strong internal validity near the cutoff but weaker external validity for students with scores far from 80.

Optional R-Based Regression Exercises

The exercises in this section are optional enrichment for students who want to practice linear regression with R. They are not part of the required problem set unless explicitly assigned by the instructor.

A. Election and Conditional Cash Transfer Program in Mexico

In this exercise, we analyze data from studies that estimate the electoral impact of Progresa, Mexico's conditional cash transfer program (CCT program).¹ The original study relied on a randomized evaluation of the CCT program in which eligible villages were randomly assigned to receive the program either 21 months (early Progresa) or 6 months (late Progresa) before the 2000 Mexican presidential election. The author hypothesized that the CCT program would mobilize voters, leading to an increase in turnout and support for the incumbent party (PRI, or Partido Revolucionario Institucional).

The data are available as `progres.csv`. Each observation is a precinct. Table 1 describes the main variables.

Table 1: Conditional Cash Transfer Program (Progresa) Data

Variable	Description
<code>treatment</code>	whether a precinct contains an early-Progresa village
<code>pri2000s</code>	PRI votes in 2000 as a share of population above 18
<code>pri2000v</code>	official PRI vote share in the 2000 election
<code>t2000</code>	turnout in 2000 as a share of population above 18
<code>t2000r</code>	official turnout in the 2000 election
<code>pri1994</code> , <code>pan1994</code> , <code>prd1994</code>	party vote counts in 1994
<code>pri1994s</code> , <code>pan1994s</code> , <code>prd1994s</code>	party vote shares relative to adult population in 1994
<code>pri1994v</code> , <code>pan1994v</code> , <code>prd1994v</code>	official party vote shares in 1994
<code>t1994</code> , <code>t1994r</code>	1994 turnout measures
<code>votos1994</code>	total votes cast in the 1994 presidential election
<code>avgpoverty</code>	precinct average of village poverty index
<code>pobtot1994</code>	total population in the precinct
<code>villages</code>	number of villages in the precinct

1. Estimating the impact of the CCT program.

- Import `progres.csv`.
- Compare average turnout and PRI support between treated (early Progresa) and control (late Progresa) precincts.
- Regress turnout and PRI support rates (`t2000` and `pri2000s`) on `treatment`.
- Interpret and compare the difference-in-means and regression results. Do the results support the hypothesis that the program mobilized voters and increased PRI support?

2. Adding pretreatment covariates.

¹Based on Ana de la O (2013), "Do conditional cash transfers affect voting behavior? Evidence from a randomized experiment in Mexico," and Kosuke Imai, Gary King, and Carlos Velasco (2015), "Do nonpartisan programmatic policies have partisan electoral effects? Evidence from two large scale randomized experiments."

- (a) Fit a linear regression model for each outcome (`t2000` and `pri2000s`) including `avgpoverty`, `pobtot1994`, `votos1994`, `pri1994`, `pan1994`, and `prd1994`.
 - (b) Compare these results to the previous question. Do the estimated program effects change?
 - (c) Explain why these variables are called pretreatment covariates and why that matters for causal interpretation.
- 3. Alternative model specifications.**
- (a) Replace count predictors with their share-based equivalents: `t1994`, `pri1994s`, `pan1994s`, and `prd1994s`.
 - (b) Apply a natural-log transformation to `pobtot1994`.
 - (c) Include `avgpoverty` as an additional predictor and compare the results. Which model fits the data better?
- 4. Examining balance of pretreatment variables.**
- (a) Use box plots to compare treatment and control groups for `pobtot1994`, `avgpoverty`, `t1994`, and `pri1994s`.
 - (b) Comment on the patterns observed in the box plots.
 - (c) Explain why balance in pretreatment variables matters in a randomized evaluation.
- 5. Using official turnout and vote shares.**
- (a) Replace the outcome variables with official turnout (`t2000r`) and official PRI vote share (`pri2000v`).
 - (b) Estimate the average treatment effect using a regression with `avgpoverty`, log-transformed `pobtot1994`, and previous election variables (`t1994r`, `pri1994v`, `pan1994v`, `prd1994v`).
 - (c) Briefly interpret the results.
- 6. Exploring heterogeneous treatment effects.**
- (a) Fit a model with `treatment`, log-transformed `pobtot1994`, `avgpoverty`, `avgpoverty`², and interactions between `treatment` and the poverty terms.
 - (b) Estimate the treatment effect at different levels of `avgpoverty`.
 - (c) Plot the estimated effects and comment on the pattern.

B. Government Transfer and Poverty Reduction in Brazil

A central question in public policy is whether giving more resources to local governments improves citizens' lives. One view is that additional public money may be wasted or captured by local elites, especially where institutions are weak. Another view is that additional resources may improve public services when citizens strongly need them and politicians face pressure to respond.

This exercise uses a regression discontinuity design to study increased government transfers in Brazil.² Until 1991, the formula for government transfers to individual Brazilian municipalities depended partly

²Based on Stephan Litschig and Kevin M. Morrison (2013), "The impact of intergovernmental transfers on education outcomes and poverty reduction."

on municipality population. Municipalities just above certain population thresholds received a larger transfer coefficient than municipalities just below them. The data are available as `transfer.csv`. Table 2 describes the main variables.

Table 2: Brazilian Government Transfer Data

Variable	Description
pop82	population of the municipality in 1982
poverty80, poverty91	poverty rate in 1980 and 1991
educ80, educ91	average years of education in 1980 and 1991
literate91	literacy rate in 1991
state	state where the municipality is located
region	region where the municipality is located
id	municipal ID
year	year of measurement

In this exercise, focus on the first three relevant cutoffs: 10,188, 13,584, and 16,980 inhabitants. The RDD compares municipalities located very close to these thresholds.

1. Regression discontinuity design assumptions.

- (a) State the required identifying assumption for the regression discontinuity design.
- (b) Interpret the assumption in this specific application.
- (c) Describe a scenario where the assumption would be violated.
- (d) Discuss the advantages and disadvantages of using this design here.

2. Creating a variable for proximity to the cutoff.

- (a) Calculate the difference between each municipality's population and the closest cutoff.
- (b) Standardize this measure by dividing the difference by the corresponding cutoff and multiplying by 100.
- (c) Interpret the resulting normalized proximity variable.

3. Estimating causal effects near the cutoff.

- (a) Subset the data to include only municipalities within 3 percentage points of the funding cutoff on either side.
- (b) Use regressions to estimate the effect of government transfers on educational attainment, literacy, and poverty.
- (c) Provide a brief substantive interpretation of the results.

4. Visualizing the analysis.

- (a) Create a plot of the data points, fitted regression lines, and the population threshold.
- (b) Briefly comment on the patterns observed in the plot.

5. Difference-in-means analysis.

- (a) Compute the difference in means for educational attainment, literacy, and poverty between municipalities above and below the threshold.
- (b) Compare these estimates to the regression estimates from question 3.
- (c) Discuss whether this analysis requires the same assumption as the RDD regression.
- (d) Evaluate which estimates are more appropriate and why.

6. Varying the analysis window.

- (a) Repeat the regression analysis from question 3, varying the analysis window from 1 to 5 percentage points above and below the threshold.
- (b) Obtain the estimate for each window width.
- (c) Briefly comment on how the results change as the window changes.

7. Validity check using pretreatment data.

- (a) Conduct the same analysis as in question 3, but use poverty and education measures from 1980, before the population-based government transfers began.
- (b) Interpret the results and evaluate what they imply for the validity of the RDD analysis.